Course Name: Physics for Quantum Computing
Course Code: TEE7251
Faculty: Engineering
Course Credit: 3
Course Level: 1
Sub-Committee (Specialization): Applied Sciences
Learning Objectives:

Describe fundamentals of quantum mechanics, particle in a box and quantization of energy levels and Dirac notations.
Describe the concept of energy levels in solids, E-k energy diagrams, and develop the ability to estimate the probability of energy level occupancy.
Explain the concepts of intrinsic and extrinsic semiconductors and familiarize applications of semiconductors.
Develop the understanding of light-matter interactions and describe applications thereof such as lasers.
Describe properties of superconductors and explain characteristic length scales related to superconductors.
Explain the working principle or fundamental theory behind the given experiment in the Physics Lab.
Demonstrate the given experiment in the Physics lab. Explain data analysis of the recorded data. Explain calculations or presentation of the acquired data.

Books Recommended:

Book	Author	Publisher
Quantum Computation and Quantum Information	M. A. Nielsen & I.Chuang	Cambridge University Press
Lecture notes	Prof. John Preskill	California Institute of Technology
An introduction to Quantum Computing	Phillip Kaye, Raymond Laflamme et. al.,	Oxford University press
Quantum Computing for Everyone	Chris Bernhardt	The MIT Press,Cambridge
Quantum Computing Explained Wiley Interscience	David McMahon	Wiley Interscience, IEEE Computer Society
Quantum Mechanics Concepts and Applications	Nouredine Zet	Edison Wiley India Pvt. Ltd.
Introduction to Quantum Mechanics	Griffiths, D. J.,	Pearson Prentice Hall,
Solid state Physics	S. O. Pillai	New Age Internatinal Publications
Semiconductor Physics and Devices	Donald A Neamen	McGraw Hill Publications
Electronic and Optoelectronic properties of Semiconductors	J Singh	Cambridge University Press

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	**Introduction to Origin of Quantum mechanics**	**5**	**5**

	Wave nature of Particles, de-Broglie wavelength, Time-dependent and time independent Schrodinger equation. Born interpretation, Particle in 1-D box, probability current, Expectation values.		
2	**Lab Assignments** Determination of reaction time and error analysis methods To study the Photoelectric effect. To determine the first excitation potential of the given gas by Frank-Hertz experiment	**8**	**4**
3	**Linear algebra for quantum mechanics** Linear algebra for quantum mechanics, operators and matrices, Dirac Notation, Stern-Gerlach experiment, Bloch sphere, Entanglement Bits and Qubits. Qubit operations, Introduction to quantum GATES. Introduction to quantum cryptography and quantum information theory	**10**	**10**
4	**Introduction to Free electron Theory** Introduction to Free electron Theory, Density of states and Origin of energy bands in solids. E-k diagram, direct and indirect bandgaps, Occupation probability, Fermi level, effective mass, phonons, Types of electronic materials: metals, semiconductors, and insulators.	**6**	**6**
5	**Lab Assignments** To study the characteristics of a solar cell and calculate the fill factor To determine the band gap of a semiconductor	**6**	**3**
6	**Introduction to semiconductors** Introduction to semiconductors, Carrier generation and recombination, Hall effect and its applications. Light interactions with matter (semiconductor), absorption, spontaneous emission, Stimulated Emission and scattering, Semiconductor Lasers (direct and indirect bandgap), phonon participation, Radiation in semiconductors, near band-gap radiative transitions and applicationsIntroduction to superconductivity and its applications in Quantum computers.	**9**	**9**
7	**Lab Assignments** To determine the wavelength of components of mercury spectrum using diffraction grating To determine the wavelength of Laser using diffraction grating To determine the angular divergence of a laser source. To study the I-V characteristic curves of semiconducting diodes To determine the velocity of ultrasonic waves using ultrasonic interferometer To determine the velocity of sound waves using Kundts tube method.	**16**	**8**
	Total	**60**	**45**

Pre Requisites:

NA

Evaluation:

Assignment
Class test
Quiz
Viva
Exercise
End Sem Examination

Pedagogy:

Classroom Teaching
Flipped Classroom
Presentations
Lab Practice

Expert:

Dr. Pramod Kumar Pandey,Assistant Professor,SIT, Nagpur